INCH-POUND

MIL-PRF-23199/1D(SH)
29 September 2023
SUPERSEDING
MIL-PRF-23199/1C(SH)
4 April 2017

PERFORMANCE SPECIFICATION

PACKAGING, PACKING AND MARKING (PP&M) REQUIREMENTS FOR INSTRUMENTATION AND CONTROL EQUIPMENT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Purpose. This specification covers the requirements for packaging, packing, and marking of Instrumentation and Control components to safeguard cleanliness, prevent damage and corrosion, and preserve identification during shipment and storage.

1.2 Applicability. This specification is a stand-alone specification that is specific to I&C equipment, and therefore, it is not necessary to refer to the base MIL-PRF-23199 specification. Previous revisions to MIL-PRF-23199 (i.e. MIL-P-23199) referenced levels A, B and C of packaging and packing. This specification eliminates these level designations; thus, any reference to these levels in contractual documents is not applicable. The entire specification should be assumed applicable unless otherwise noted in contractual documents.

1.2.1 Field service. Unless otherwise specified or requested by the Government Procurement Agency (GPA), the requirements contained herein do not apply to materials or equipment that is hand carried by supplier field service personnel. The Seller field service personnel are responsible for protecting and handling the hand carried items. Materials and equipment that is shipped separately in support of field service is required to be packaged, packed and marked in accordance with this specification.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or e-mailed to commandstandards@navy.mil, with the subject line “Document Comment”. Since contracting information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A PACK-2023-005

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1.2.2 Shipment in place. Unless otherwise specified or requested by the GPA, the requirements contained herein do not apply to items where the GPA has authorized shipment in place. In such cases, the Seller is responsible for storing, protecting and maintaining the identification of the items until the GPA has authorized shipment to final destination. Items shipped in place are considered Government Furnished Equipment (GFE) and shall be controlled as such in accordance with Seller procedures for control of GFE. The requirements specified herein apply for shipment to final destination.

1.2.3 Specialized, spare, and miscellaneous components. Specialized, spare or miscellaneous components that do not fall within the standard packaging requirements described within MIL-PRF-23199/1, shall be packaged in accordance with MIL-STD-2073 and any other special packaging requirements invoked by the GPA.

1.3 PP&M procedures. Seller PP&M procedures written in accordance with the requirements contained herein are subject to GPA on-site review. The Seller may also submit the procedure via the Coordinated Procedure Review System (CPRS). Alternatively, the Seller may develop a facility based PP&M plan that describes, in general terms, the methods to be used for PP&M. The plan must document compliance with the requirements specified herein. If this method is chosen, PP&M details such as lumber size, etc. may be implemented in Seller work instructions. Such details and the overall plan shall be made available to the GPA for on-site review upon request. Deviations from this specification may be authorized by the GPA on a case basis (i.e., alternate packaging methods may be authorized when the packaged equipment is intended for near term installation and long term preservation is not required, or otherwise when considered acceptable by the GPA.) When the GPA authorizes commercial packaging, ASTM D3951 should be used as a guideline. However, the Seller remains responsible to safeguard cleanliness, prevent damage (including electro-static and moisture protection if required) and preserve identification during shipment. The use of “loose fill” cushioning is prohibited and all marking requirements identified in this specification remain in effect.

1.3.1 Loading and blocking. The Seller shall prepare a procedure providing the minimum requirements for loading and blocking of each equipment type. One generic facility wide plan is acceptable. Loading and blocking requirements shall apply for all equipment packaged in accordance with 3.2.1 or 3.2.2. The procedure shall be made available for GPA on-site review. All shipments shall be made in closed trailers. Loading and blocking shall be in accordance with DOT CFR Title 49 392.9 (Safe Loading), 393.100 (General rules for protection against shifting or falling cargo), 393.102 (Securement systems), and 393.104 (Blocking and bracing).

1.4 Item Unique Identification (UID). The assignment and marking of Item Unique Identification (UID) does not apply to equipment packaged per the requirements of MIL-PRF-23199/1.

1.5 Tamper-Evident Packaging (TEP). Tamper-Evident Packaging (TEP) of components shall be delivered to the GPA in accordance with the applicable revision of AOR-TEP as identified on the purchase order.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification, or recommended for additional information, or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that all

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specified requirements documents cited in sections 3, 4, or 5 of this specification must be met, whether or not each is listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

PPP-C-795	-	Cushioning Material, Packaging (Flexible Closed Cell Plastic Film, For Long Distribution Cycles)
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

COMMERICAL ITEM DESCRIPTION

A-A-3129	-	Cushioning Material, Flexible Open Cell Plastic Film (For Packaging Applications)
A-A-59135	-	Packaging Material, Sheet

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-117	-	Bags Heat-Sealable
MIL-I-8835A	-	Indicator, Humidity, Card, Chemically Impregnated
MIL-DTL-17361	-	Circuit Breakers, Air, Electric, Insulated Housing, (Shipboard Use), General Specification for
MIL-DTL-17587	-	Circuit Breakers, Low Voltage, Electric Power, Air, Open Frame, Removable Construction
MIL-DTL-24466	-	Polyethylene Bags, Sheet, and Tubing, Green
MIL-PRF-81705	-	Barrier Materials Flexible, Electrostatic-Free, Heat Sealable
MIL-DTL-81997	-	Pouches, Cushioned, Flexible, Electrostatic-Free Reclosable, Transparent
MIL-D-3464	-	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-PRF-83671	-	Foam-In-Place Packaging Materials

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-2041	-	Control of Detrimental Materials
MIL-STD-2073-1	-	Standard Practice for Military Packaging
MIL-STD-147	-	Standard Practice for Palletized Unit Loads

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-773	-	Electrostatic Discharge Protective Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil or https://assist.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title 48, Volume 3, Chapter 2, Part 245 Subpart 245.7001-2	-	Federal Acquisitions Regulations System Defense Acquisitions System, Department of Defense Government Property DD-1149, Requisition and Invoice Shipping Document
Title 48, Volume 3, Part 252, Subpart 252.246-7000	-	Solicitation Provisions and Contract Clauses DD-250, Material and Inspection Report
Title 49, Volume 5, Chapter 3, Parts 390-399	-	Transportation Federal Highway Administration Federal Motor Carrier Safety Regulations

(Copies of these documents are available online at http://www.ecfr.gov.)

DEPARTMENT OF DEFENSE MANUAL

DoD 4140.65	-	Department of Defense Manual, Compliance for Defense Packaging: Phytosanitary Requirements for Wood Packaging Material (WPM)
DoD 5220.22	-	National Industrial Security Program Operating Manual (NISPOM)

(Copies of these documents are available online at http://www.DTIC.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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IPC ASSOCIATION CONNECTING ELECTRONICS INDUSTRIES (IPC)

IPC/JEDEC J-STD-033	-	Handling, Packing, Shipping and Use of Moisture, Reflow, and Process Sensitive Devices

(Copies of these documents are available online at https://www.ipc.org.)

ASTM INTERNATIONAL (ASTM)

ASTM D3951	-	Standard Practice for Commercial Packaging
ASTM D6251	-	Standard Specification for Wood-Cleated Panelboard Shipping Boxes
ASTM D7478	-	Standard Specification for Heavy Duty Sheathed Wood Crates
ASTM D5118	-	Standard Practice for Fabrication of Fiberboard Shipping Boxes

(Copies of these documents are available from http://www.astm.org.)

INTERNATIONAL PLANT PROTECTION CONVENTION (IPPC)

ISPM No. 15	-	Regulation of Wood Packaging Material in International Trade

(Copies of this document are available online from the International Plant Protection Convention, at https://www.ippc.int)

ELECTRONICS COMPONENTS INDUSTRY ASSOCIATION (EIA)

ANSI/EIA-481	-	4 mm Through 200 mm Embossed Carrier Taping and 8 mm & 12 mm Punched Carrier Taping of Surface Mount Components for Automatic Handling

(Copies of this document are available online at https://www.ecianow.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Materials.

3.1.1.1 Fiberboard boxes. Fiberboard boxes shall be in accordance with ASTM D5118 Type CF, Class weather resistant, corrugated are acceptable. The gross weight of a fiberboard

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box, including contents, shall not exceed the manufacturer's maximum gross weight described on the box. In no case, shall the gross weight of a fiberboard box including contents exceed 60 pounds.

3.1.1.2 Cleated, palletized, panelboard boxes. Cleated, palletized, panelboard boxes shall be in accordance with ASTM D6251, Type I or Type II, Class 1 or Class 2, Style A are acceptable. Palletization shall be in accordance with MIL-STD-147. The gross weight of a panelboard box, including contents, shall not exceed the manufacturer's maximum gross weight described on the box. In no case, shall the gross weight of a panelboard box including contents exceed 200 pounds.

3.1.1.3 Wooden boxes. Wooden boxes for gross shipments under 200 pounds do not require skids. Component shipments in wooden boxes exceeding 200 pounds shall have skids suitable for fork lifting. Wooden boxes shall be girth strapped using rust resistant steel banding around the box in not less than two places. For skids, the girth straps shall be applied to and immediately adjoining the inner edge of each skid. For skid-type base boxes, wooden blocking shall be affixed to the base to prevent fork-lifting of loads from the ends. The blocking shall not extend beyond the sheathing envelope, shall be permanently attached, and shall not interfere with sling handling of the load. Wooden boxes, including contents, shall not exceed 800 pounds. Reinforced wooden box sketches that include the blocking, bracing and anchoring arrangement shall be provided in the Seller's procedure.

3.1.1.4 Wooden crates. Components exceeding 800 pounds shall be packed in skid based wooden crates per the construction requirements of ASTM D7478 using material per 3.1.1.5. Wooden blocking shall be affixed to the base to prevent fork-lifting of loads from the ends. The blocking shall not extend beyond the sheathing envelope, shall be permanently attached, and shall not interfere with sling handling of the load. Reinforced wooden crate sketches that include the blocking, bracing and anchoring arrangement shall be provided in the Seller's PP&M procedure.

Unless otherwise specified by the GPA, for items packed in accordance with 3.2.1.2, the front of the crate shall be fastened to the sides (ends) using a minimum number of quick release fasteners, lag bolts or deck screws. A minimum number of these types of fasteners shall also be used to fasten the bottom of the sides (ends) to the skid base. Except for attachment of the quick release fasteners, nails, staples, or other types of fasteners, shall not be used to secure the front of the crate to the sides (ends) of the crate, or the sides (ends) of the crate to the skid base.

3.1.1.5 Materials of construction. Wooden boxes and crates shall be enclosed and constructed using commercial quality lumber and/or plywood fabricated in a manner to adequately protect the component. Knots, knotholes, splits and moisture content shall not impair the integrity of the container and component. Commercial nails and/or other fasteners shall be of sufficient size, spacing and quantity to provide a construction that will adequately protect the component under normal handling during shipment. Nails and other fasteners shall be driven so that neither the head nor the point projects above and through the surface of the wood (except for the quick-release fasteners and lag bolt heads defined in par 3.1.1.4). Steel banding shall be rust resistant.

3.1.1.5.1 Wood packaging materials. Wood packaging materials (WPM) shall be in accordance to the latest revision of DoD 4140.65 and ISPM No. 15 for all shipments. Assets must comply when packed in or on wooden pallets, skids, load boards, pallet collars, wooden

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boxes, reels, dunnage, crates, frames, blocking, bracing, and cleats. The wood treatment WPM certification markings must be easily visible to inspectors.

3.1.1.6 Envelopes for use with detectors and sensors. Envelopes shall be pigmented green polyethylene film conforming to the following requirements:

(a) The polyethylene shall meet the detrimental material control limits in MIL-STD-2041 for mercury and leachable halides and sulfur. Polyethylene meeting the requirements of MIL-DTL-24466 is considered acceptable.

(b) The polyethylene shall be free from gels, holes and foreign material and have uniform texture.

(c) The polyethylene shall be capable of meeting the physical properties for Type II polyethylene as described in A-A-3174.

(d) Envelope size shall be kept to a minimum.

(e) Envelope nominal thickness shall be 0.006 inches (0.004 inches minimum). Multiple layers may be used for component protection.

3.1.1.7 Electrostatic discharge (ESD) sensitive item control. All ESD sensitive items subject to damage by electromagnetic and electrostatic field forces shall be handled at an approved field force protective workstation (i.e. ESD protected workstation). Informational guidance for ESD control and packaging is detailed in MIL-HDBK-773.

3.1.1.7.1 ESD packaging equipment and materials. All handling equipment and materials (e.g. wrist straps, bags, boxes, foams, trays, totes, etc.) used for packaging ESD sensitive items shall be purchased directly from the original manufacturer or their authorized distributors.

3.1.1.7.2 ESD packaging. ESD sensitive items shall be packaged by initially wrapping and/or cushioning, and then sealing in a unit bag using the materials identified in Table I below. Reclosable cushioned pouches conforming to MIL-DTL-81997 Type I or Type II may be used in lieu of an initial wrap or cushioning.

Only antistatic or conductive materials are allowed inside the unit pack ESD bag. The desiccant and humidity indicator shall be attached to the outside of an inner bag, cushion, or wrap. However, the desiccant and humidity indicator shall not come in contact with the ESD sensitive item being packaged.

Lead or terminal configurations for all items shall be protected and maintained as manufactured without causing loads or stresses capable of causing damage to the item. Materials used to maintain item position and lead or terminal configuration shall permit item removal without damage to the item.

ESD tape used inside the unit package bag shall contain an ESD symbol pattern.

ESD sensitive caution labels shall be applied in accordance with MIL-STD-129.

Connectors and plugs (power or signal) shall be covered with conductive caps, plugs or similar means, and marked with the MIL-STD-129 ESD symbol pattern.

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TABLE 1. ESD packaging material process and material specifications

Process	Material to conform to
Initial Wrap or Cushioning	MIL-PRF-81705 Type I or Type III or A-A-59135 Class 2, Grade B or Grade D or A-A-3129 Grade B or PPP-C-795 Class 2 or Class 3, or PPP-C-1797 Type II or bags conforming to MIL-DTL-117, Type I, Type II, Type III or Type IV, Class F or Class H, Style 1, Style 2 or Style 3.
Cushioned Pouches	MIL-DTL-81997 Type I or Type II
Unit Bag	MIL-DTL-117, Type I, Type II or Type IV, Class F, Style 1, Style 2 or Style 3.
Desiccant	MIL-D-3464 Type I General Purpose, Type II Non-Dusting, or Type III For Specific Conditions (if specified in contract).
Humidity Indicator Card	MIL-I-8835, Indicator, Humidity, Card, Chemically Impregnated.

3.1.1.8 Other packaging materials. At a minimum, other packaging materials shall meet Table I material conformity requirements and shall be capable of performing their intended function. Bag material shall be able to be heat sealed resulting in a water/vapor proof barrier and shall be of sufficient thickness to prevent puncture and tearing during handling and shipment. Cushioning materials, blocking and bracing shall be applied as required to protect the item and the enclosing media and restrict the movement of the item within the container.

3.1.1.8.1 "Loose fill" cushioning. Use of "loose fill" cushioning is prohibited.

3.1.1.9 Material for plastic encapsulated microcircuit (PEM) devices. For plastic encapsulated microcircuit devices a moisture barrier bag (MBB) that meets MIL-PRF-81705 Type I requirements for flexibility, ESD protection, mechanical strength, and puncture resistance as described in IPC/JEDEC J-STD-033 shall be provided. The bag shall be evacuated (hand or vacuum) free of air and heat-sealed. The water vapor transmission rate (WVTR) shall be in accordance with IPC/JEDEC J-STD-033. Humidity indicator cards and desiccant shall be placed inside MBBs per the requirements of IPC/JEDEC J-STD-033. The desiccant and humidity indicator shall be attached to the outside of an inner bag or wrap. The desiccant shall not come in contact with the item being packaged.

3.1.2 Foreign materials. All items packaged and packed, and all packaging/packing medium shall be free from foreign object debris (FOD).

3.2. Detailed packaging, packing and marking requirements.

3.2.1 Cubicles, panels and equipment enclosures.

3.2.1.1 Packaging. Cubicles, panels and equipment enclosures shall be packaged in a heat sealed, floating bag with desiccant and humidity indicators per the preservation requirements of MIL-STD-2073-1 Method 53 (floating watervaporproof bag with desiccant, heat

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sealed). Materials shall be in accordance with 3.1. For cubicles, panels and equipment enclosures with exposed ESD sensitive components mounted directly on the enclosure, the enclosure shall be wrapped with ESD protective barrier material conforming with section 3.1.1.7.

As an alternative, a non-ESD protective barrier material may be used if the enclosure connector(s) are covered with ESD protective shipping cap(s), or other ESD protective material meeting section 3.1.1.7 ESD packaging requirements. ESD sensitive connectors and plugs (power or signal) covered with conductive caps, plugs or similar means shall be marked with a legible MIL-STD-129 ESD sensitive electronic device warning symbol (or circle with three arrows pointing toward the center of the circle if the MIL-STD-129 ESD symbol marking is not legible).

3.2.1.2 Packing. Cubicles, panels and equipment enclosures shall be packed in wooden crates (3.1.1.4) designed to be shipped and handled under cover and stored in warehouses or other structures affording equivalent protection from weather. The packing shall provide adequate protection during multiple reshipments.

3.2.1.2.1 Anchoring, blocking, bracing, cribbing, strapping, and cushioning. **Cubicles, panels, and equipment enclosures shall be anchored, blocked, braced, cribbed, and strapped inside of wooden crates (3.1.1.4) to secure and prevent any movement of the item within the container. Blocking, bracing, cribbing, and strapping (banding) shall be used inside of the shipping crate at multiple horizontal and vertical points to anchor, immobilize, stabilize and prevent movement of the the item during handling, shipping, and storage. Cushioning and padding shall be used inside of the shipping container at points where the item is anchored, blocked, braced, and strapped to protect the item. Cushioning and padding shall also be used to absorb the energy of shocks and vibrations encountered during handling, shipping, and storage of the item.**

3.2.1.2.2 Shock and tilt indicators. Commercial grade shock and tilt indicators which can detect and record rough handling or unacceptable tilting, and which provide a visual deterrent to improper handling of shipments, shall be affixed to the crate in such a manner as to provide for monitoring on all six axes (e.g., one indicator on the front and one on the adjacent side of the crate).

3.2.1.3 Marking. In addition to the marking requirements per 3.3, the following marking shall be provided:

3.2.1.3.1 Sheltered building marking. The outer container shall be marked on all four sides in red as follows:

STORAGE REQUIREMENTS: SHELTERED BUILDING
MINIMUM TEMPERATURE 40° F

3.2.1.3.2 Front location marking. The crate shall be marked to identify the location of the front of the cubicle, panel or equipment enclosure and any special handling instructions, e.g., “LIFT FROM THIS END ONLY”. For asymmetrical, off-center loads, the preferred fork-lift opening shall be marked and all other sides shall be marked “DO NOT LIFT FROM THIS END”. (See 3.1.1.3 or 3.1.1.4.)

3.2.1.3.3 Shock and tilt indicators. When shock and tilt indicators are used (see 3.2.1.2.2) the Seller shall add a note on the shipping crate adjacent to the indicator directing the Receiving

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Facility to annotate the delivery receipt, and to contact the Seller and the GPA if the shipment is received with an indicator that identifies the possibility of damage during transit.

3.2.2 Drawers and chassis.

3.2.2.1 Packaging. Drawers and chassis shall be packaged in a heat sealed bag with desiccant and humidity indicators per the preservation requirements of MIL-STD-2073-1 Method 51 or Method 52. Materials shall be in accordance with 3.1. For drawers or chassis with ESD sensitive components mounted directly on the drawer or chassis (other than plug-in modules), the barrier shall be made from ESD protective material (3.1.1.7). For drawers containing plug-in modules containing ESD sensitive components, the barrier material shall be made from ESD protective material. As an alternative, a non-ESD protective barrier material may be used if the drawer connector(s) is covered with an ESD protective shipping cap, or other ESD protective material (so that no drawer or chassis required ESD sensitive component is exposed).

3.2.2.1.1 Protection to prevent connector and pin damage. Protection shall be added to protect exposed chassis and drawer components including, but not limited to connectors and connector pins, heat sinks, shock pins, buttons, switches, knobs and other hardware from damage (i.e. breaking or bending during storage, handling, or shipment).

3.2.2.2 Packing. Drawers and chassis shall be individually packed in wooden boxes (3.1.1.3) designed to be shipped and handled under cover and stored in warehouses or other structures affording equivalent protection from weather. The packing shall afford adequate protection during multiple reshipments.

3.2.2.3 Marking. Marking shall be in accordance with 3.3.

3.2.3 Detectors and sensors (other than neutron/gamma detectors).

3.2.3.1 Packaging. Detectors and sensors (other than neutron/gamma detectors) shall be packaged in a heat sealed green polyethylene bag (3.1.1.6) with desiccant and humidity indicators per the preservation requirements of MIL-STD-2073-1 Method 51. Materials shall be in accordance with 3.1.

3.2.3.2 Packing. Detectors shall be individually packed in wooden boxes (3.1.1.3) designed to be shipped and handled under cover and stored in warehouses or other structures affording equivalent protection from weather. The packing shall afford adequate protection during multiple reshipments.

3.2.3.2.1 Viewing window requirements for resistance temperature detectors (RTDs). For RTDs, the wooden packing boxes shall have a viewing window that is a 1.0 +/- 0.25 inch radius hole centered on one of the ends. The humidity indicator, within the sealed bag, shall be affixed such that it shall be visible from the viewing window. The viewing window shall be covered by transparent thermoplastic material with a nominal thickness of 0.125 inch or greater that is firmly secured to the box with screws.

3.2.3.3 Marking. Marking shall be in accordance with 3.3.

3.2.3.4 Additional requirements for primary plant detectors. For detectors intended to be in contact with primary plant fluid, contract cleanliness requirements shall apply through all steps of packaging and handling. In addition, the green polyethylene bag shall be purged with dry

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inert gas (nitrogen or argon) having a dew point of minus 40 degrees Fahrenheit or lower prior to sealing.

3.2.3.5 Calibration data sheets. When calibration data sheets are required, three copies of the data sheets shall be placed in a heat sealed waterproof envelope and packaged with the detector. The data envelope shall be labeled with its contents, and the detector Standard Identifier (SI) and serial number. Additionally, the data envelope shall be labeled:

****
DATA SHEETS
REQUIRED FOR DETECTOR INSTALLATION
STORE WITH THE DETECTOR
DO NOT DISCARD
****

3.2.4 Neutron and gamma detectors.

3.2.4.1 Packaging.

3.2.4.1.1 Neutron detector packaging. Each neutron detector shall be packaged per the preservation requirements of MIL-STD-2073-1 Method 51. The detector shall be placed in a green polyethylene bag (3.1.1.6) with desiccant and humidity indicators. A second bag made from water/vapor proof material shall be heat sealed around the first, green polyethylene bag. The water/vapor proof bag shall be purged with dry inert gas (nitrogen or argon) having a dew point of minus 40 degrees Fahrenheit or lower prior to sealing. Materials shall be in accordance with 3.1.

3.2.4.1.2 Gamma detector packaging. Each gamma detector shall be packaged in a heat sealed green polyethylene bag (3.1.1.6) with desiccant and humidity indicators per the preservation requirements of MIL-STD-2073-1 Method 51. The green polyethylene bag shall be purged with dry inert gas (nitrogen or argon) having a dew point of minus 40 degrees Fahrenheit or lower prior to sealing. A second bag made from water/vapor proof material shall be heat sealed around the first green polyethylene bag. Materials shall be in accordance with 3.1.

3.2.4.2 Packing. Each detector shall be packed in a wooden box (3.1.1.3) designed to be shipped and handled under cover and stored in warehouses or other structures affording equivalent protection from weather. The packing shall afford adequate protection during multiple reshipments.

3.2.4.3 Marking. In addition to any special labeling or marking required by the Department of Transportation (DOT) and the marking requirements identified in 3.3, the following requirements apply:

3.2.4.3.1 Detector shipping container marking. The shipping container for each detector shall be marked “Fragile”.

3.2.4.3.2 Irradiated or radioactively contaminated detector marking. Irradiated (operated or tested in a reactor facility), or radioactively contaminated detectors to be shipped to a GPA program facility shall be marked and labeled in accordance with applicable regulations.

3.2.4.4 Shipping caps. Each detector connector shall have a weatherproof coaxial

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connector protective cap to prevent damage to the connector threads and degradation to the connector insulation resistance.

3.2.4.5 Integral cable. The detector's integral cable shall be coiled and held together with twisted wire ties or other suitable methods. The cable coil shall have a diameter such that no detrimental defects (including, but not limited to indentations or ripple-like marks) can occur in the contour of the outer sheath of the cable.

3.2.4.6 Information furnished with neutron detectors. A copy of the signed acceptance test data sheet with the serial number, and measured sensitivity value shall be transmitted in a heat-sealed waterproof envelope with each compensated ion chamber.

3.2.5 Special tools. Deleted.

3.2.6 Miscellaneous items. Miscellaneous items such as connectors, meters, resistors, capacitors, magnetic components, switches, microcircuits, semiconductors, sheet metal items, hardware, etc. shall be packaged and packed in accordance with the requirements for Spare Components (3.2.7). In addition, large transformers and similar types of items where the load is characterized by irregular shape or great density shall be packed in accordance with the requirements of 3.2.12.

3.2.6.1 Printed wiring assemblies (PWA), electronic assemblies, circuit card assemblies (CCA) and electronic module assemblies (other than drawers or chassis covered by 3.2.2). PWA ESD control requirements are defined in 3.1.1.7 and MIL-STD-2073-1. PWA Moisture protection requirements are defined in 3.2.7.1.6. PWA ESD handling requirements are defined in 3.2.7.1.4.4. Figures 1 and 2 show typical PP&M arrangements for these types of equipment.

Each assembly or module shall be individually unit packaged in a heat-sealed bag with desiccant and humidity indicators per the preservation requirements of MIL-STD-2073-1 Method 51 or Method 52.

3.2.6.1.1 Packaging protection to prevent connector and pin damage. Protection shall be provided to protect exposed connectors and connector pins, stab-connect feed-through terminal blocks, alignment or shock pins, and/or other protruding component(s) from damage due to breaking or bending during storage, handling, or shipment.

3.2.6.1.2 Packing. Each assembly shall be unit packed. The packing shall afford adequate protection during multiple reshipments. The unit pack (i.e., outside container) shall meet the requirements of 3.1.1.1, 3.1.1.2 or 3.1.1.3 as applicable.

3.2.6.1.3 Marking. Unit package marking is only required on the outer bag. In addition to marking requirements specified in 3.3, the unit packaging shall be marked in red ink in accordance to MIL-STD-129 Method 50 identification marking: "PACKAGED WITH DESICCANT DO NOT OPEN UNTIL READY FOR USE OR INSPECTION".

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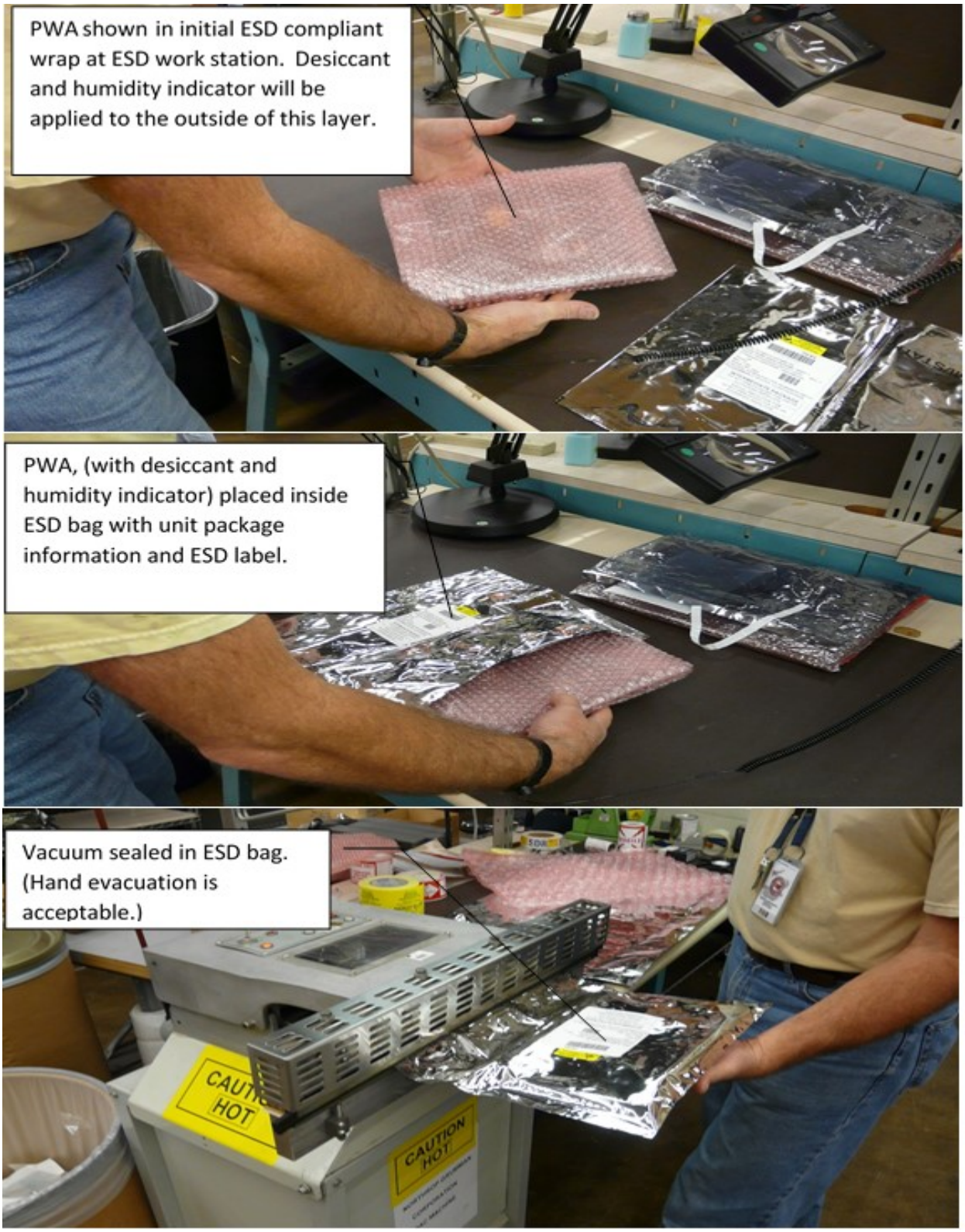


FIGURE 1. Typical PP&M sequence for a PWA or CCA.

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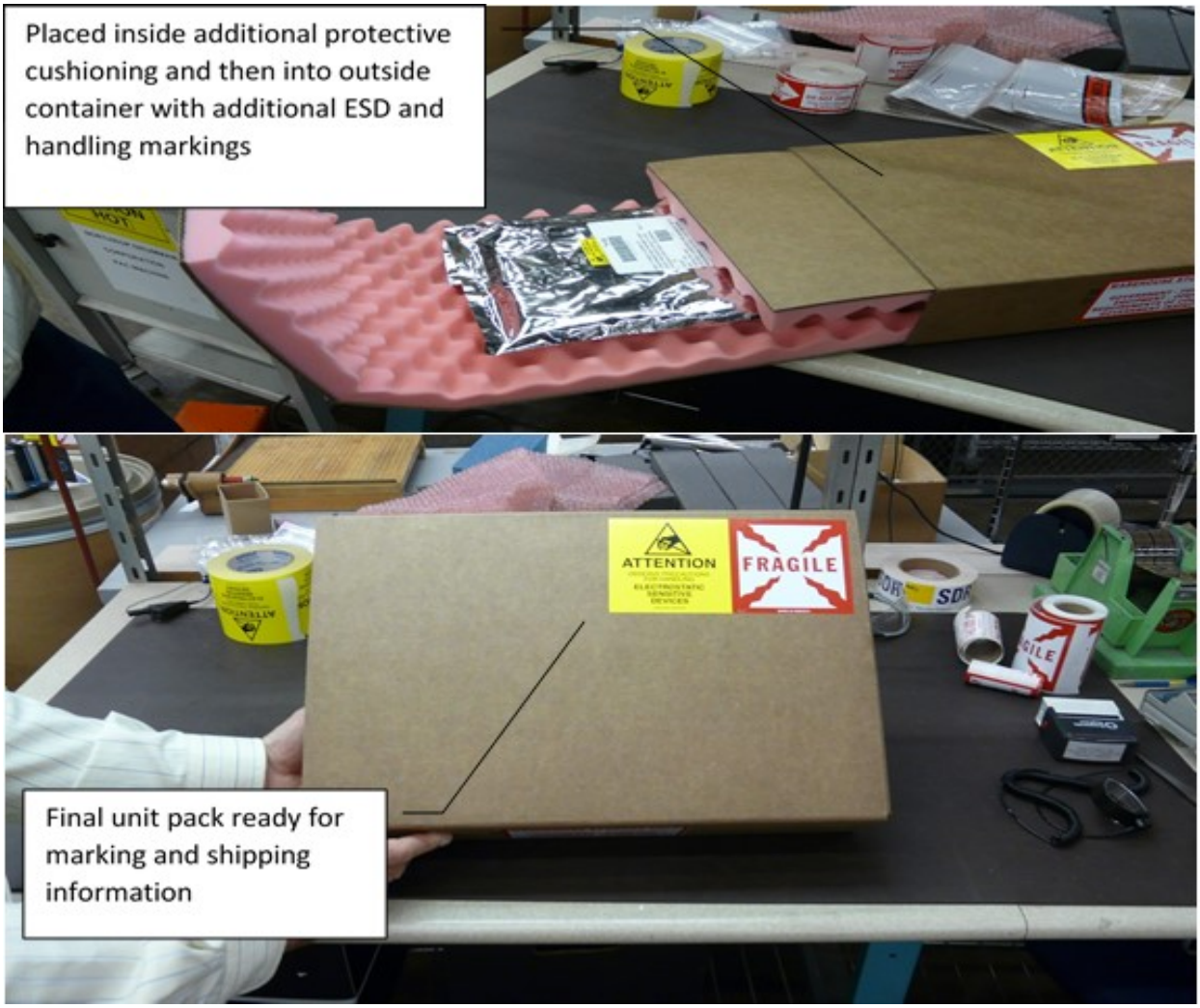


FIGURE 1. Typical PP&M sequence for a PWA or CCA − Continued.

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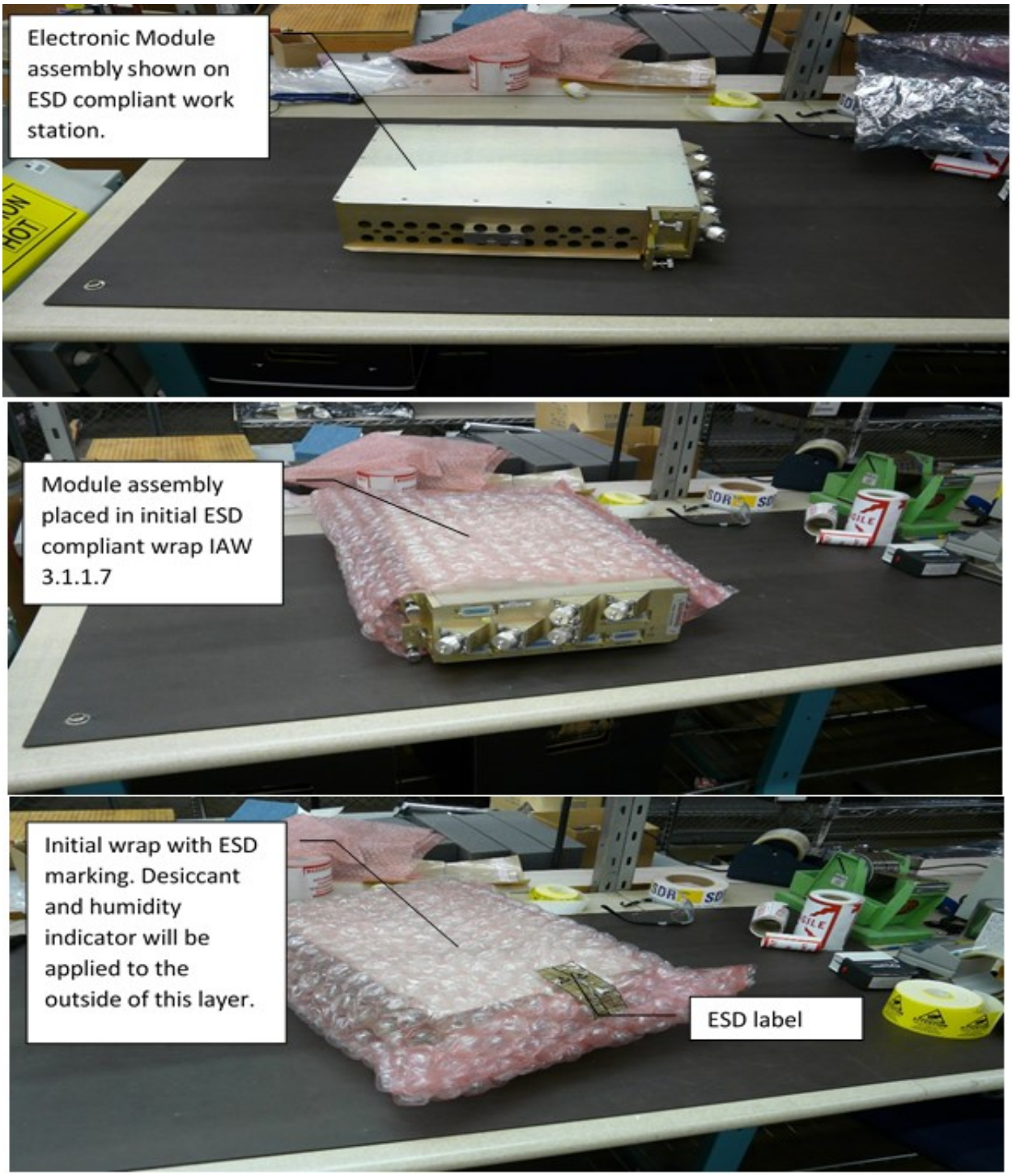


FIGURE 2. Typical PP&M sequence for an I&C electronic module assembly.

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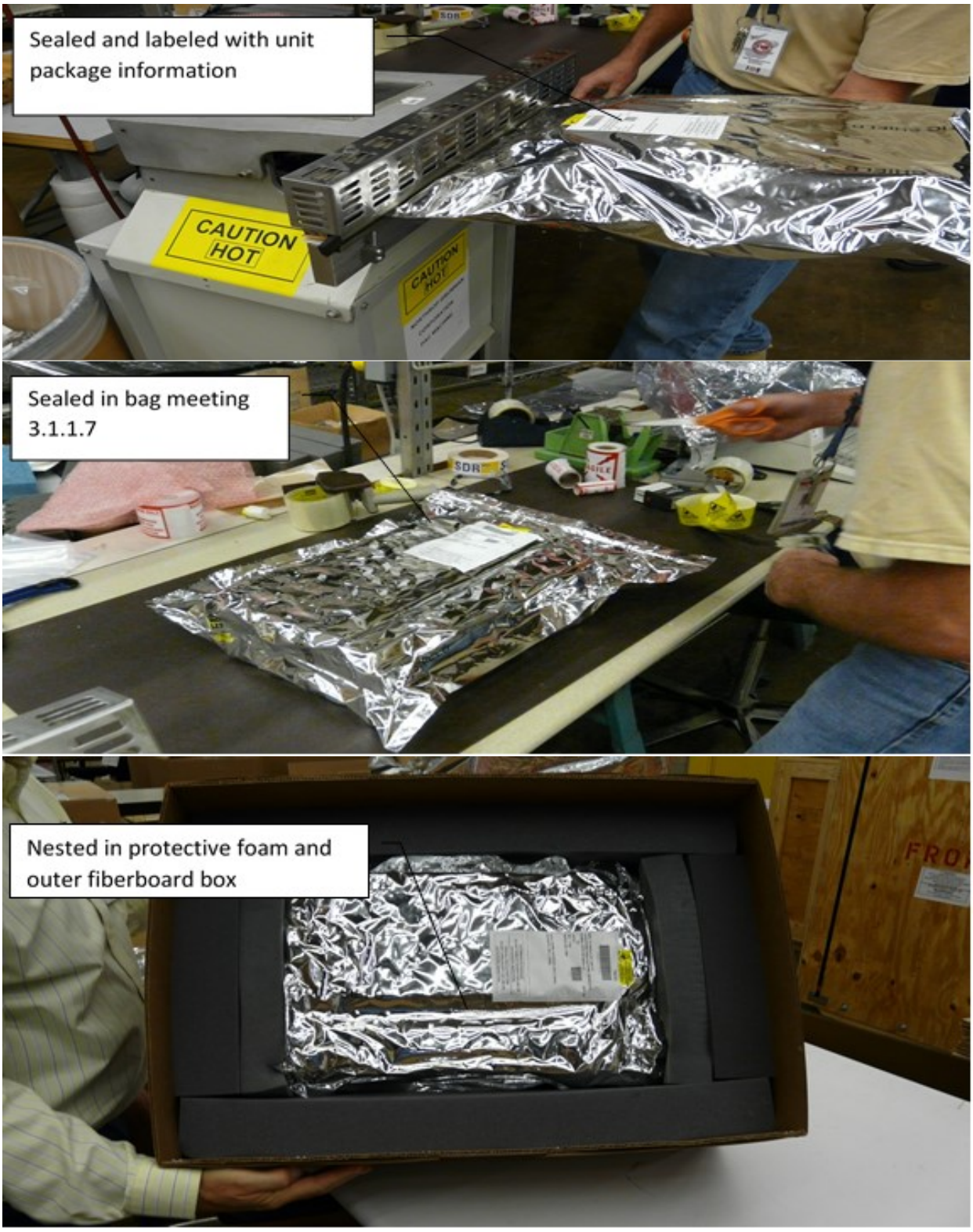


FIGURE 2. Typical PP&M sequence for an I&C electronic module assembly − Continued.

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FIGURE 2. Typical PP&M sequence for an I&C electronic module assembly – Continued.

3.2.6.2 Surface mount devices (SMD) – trays, reels (tape and reel), and tubes. This section provides requirements for lot marking, pack and packaging of surface mount electronic components that are shipped from the original manufacturer in trays, on reels, or in tubes. ESD handling requirements are defined in 3.2.7.1.4.4. The time that components are exposed to the air shall be kept to a minimum and shall be in accordance with IPC/JEDEC J-STD-033.

3.2.6.2.1 Packaging of parts in trays. This paragraph only applies to components that are provided by the component manufacturer in trays.

3.2.6.2.1.1 Reuse of manufacturer's trays. The component manufacturer's trays may be reused provided the trays meet the requirements of paragraph 3.2.6.2.1.3.

3.2.6.2.1.2 Inspection of components in trays. Each tray of components shall be inspected to assure that the tray contains the correct components, the components are in satisfactory condition and the quantity per tray (as specified in the purchase order) is correct.

3.2.6.2.1.3 Appropriate tray utilization. The Seller shall assure that the tray utilized is appropriate for the components. The tray shall support the components such that the components cannot move within the tray once a tray cover is installed. The tray shall support

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the components such that the component leads (pins/balls) are protected and not in contact with the tray.

3.2.6.2.1.4 Unit package of trays. Each tray of components shall be a unit package and an appropriate tray cover shall be provided such that the components are protected from movement within the tray. The tray covers and the tray shall be tightly strapped using a minimum of four straps; minimum two widthwise and minimum two lengthwise. The straps shall be either:

(a) static dissipative/conductive thermally sealed polypropylene or

(b) static dissipative/conductive polypropylene hook and loop (e.g., Velcro ®) strap with added D-ring.

(c) or equivalent to a) or b).

3.2.6.2.2 Packaging of parts on reels (tape and reel). This paragraph only applies to components that are provided by the component manufacturer on reels.

3.2.6.2.2.1 Sample inspection. The number of components per reel shall be as specified in the purchase order. Each reel of components shall be sample inspected (one location at the start of the reel, three separate places in the middle of the reel, and one place near the end of the reel) to assure that the reel contains the correct components and that the components are in satisfactory condition.

3.2.6.2.2.2 Subdividing components on reels. If the components received in a quantity per reel that differs from that indicated in the purchase order, then the reel of parts shall be subdivided to meet the quantity per reel identified in the purchase order.

3.2.6.2.2.3 Reels for automated handling equipment. The reels utilized shall be made specifically for use in automated handling equipment per the requirements of EIA-481. The reels shall be made of a conductive material (anti-static reels that are non-conductive are not acceptable). The hub diameter shall be the same (or larger) as the hub diameter utilized by the component manufacturer. The width of the reel shall be of the same as the width utilized by the component manufacturer. The parts shall be transferred to the new reel per the requirements of applicable document EIA-481.

3.2.6.2.2.4 Tape for mounting reels of parts used in automated equipment. All reels of parts being supplied shall have the parts mounted on the tape for use in automated equipment. The tape (with parts mounted) shall be continuous and shall meet the requirements of applicable document EIA-481 (including the requirements of minimum lead, minimum trailer, and cover tape over the entire tape including lead and trailer).

3.2.6.2.3 Packaging of parts in tubes. This paragraph only applies to components that are provided by the component manufacturer in tubes.

3.2.6.2.3.1 Components shall remain in tubes. The components shall remain in the tubes as provided by the component manufacture.

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3.2.6.2.3.2 Inspection of components in tubes. Each tube of components shall be inspected to assure that the tube contains the correct components, the components are in satisfactory condition and the quantity per tube (as specified in the purchase order) is correct.

3.2.6.2.4 Dry pack in moisture barrier bag. This paragraph provides specifications for dry packing components in moisture barrier bags.

3.2.6.2.4.1 Dry packing components stored in trays, reels, and tubes. Each tray/reel/tube of components shall be dry packed per the requirements of applicable document IPC/JEDEC J-STD-033. This includes the requirements for moisture barrier bags, desiccant and humidity indicators.

3.2.6.2.4.2 Moisture barrier bags for all components. For all components, regardless of the moisture sensitivity of the part being packaged, the moisture barrier bags shall meet the requirements intended for parts with a moisture sensitivity level of 5a as defined in IPC/JEDEC J-STD-033.

3.2.6.2.4.3 Desiccant and humidity indicators. The desiccant and humidity indicators shall be placed in the bag in a manner such that they do not come in direct contact with the components. ESD foam, board or equivalent ESD materials may be placed in the bag when necessary to assure that the desiccant and humidity indicators do not come in direct contact with the components.

3.2.6.2.4.4 Adhesive tape. Adhesive tape shall not be used inside the moisture barrier bag.

3.2.6.2.4.5 Components requiring dry packing. Components classified as level 2a through 5a shall be dried prior to sealing if the manufacturer’s exposure time (MET) is exceeded. See applicable document IPC/JEDEC J-STD-033 dry packing requirements for the detailed requirement.

3.2.6.2.4.6 Nitrogen fill of moisture barrier bag. Prior to sealing the moisture barrier bag, the bag shall be nitrogen filled, followed by evacuation of excess nitrogen/air (by hand or vacuum).

3.2.6.2.5 Fiberboard box. Each unit pack moisture barrier bag containing the tray/reel/tube of components shall then be packaged in an individual fiberboard box (unit pack box) of minimal size. Cushioning shall be provided to protect the reel/tray/tube from moving within the fiberboard box. Loose fill cushioning shall not be used (bubble wrap or closed cell foam wrap is acceptable).

3.2.6.2.6 Marking.

3.2.6.2.6.1 Marking of unit pack boxes and unit pack moisture barrier bags. Both unit pack boxes and the unit pack moisture barrier bags shall be marked per the requirements of 3.3. The quantity and unit measure marked on the unit packs shall read “1 Lot”. The description shall include the part nomenclature and in parentheses the quantity of parts per tray/reel/tube. For example: “Microcircuits (24 Pieces per Tray)” or “Microcircuits (100 Pieces per Reel)”.

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3.2.6.2.6.2 Additional markings on moisture barrier bags. The moisture barrier bag shall additionally be marked with:

(a) The standard ESD symbol per MIL-STD-129 (triangle with hand symbol inside), and

(b) In yellow, MIL-STD-129 unit pack symbol and label “ATTENTION STATIC SENSITIVE DEVICES - HANDLE ONLY AT STATIC SAFE WORK STATIONS”, and

(c) In red, MIL-STD-129 Method 50 marking “PACKAGED WITH DESICCANT – DO NOT OPEN UNTIL READY FOR USE OR INSPECTION”.

3.2.6.2.6.3 Additional markings for moisture sensitive components. For moisture sensitive components (as identified by the component manufacturer in the original, as-received packaging), the moisture barrier bag shall additionally be labeled in accordance with applicable document IPC/JEDEC J-STD-033 label requirements for the appropriate device dry pack level.

3.2.6.2.6.4 Additional markings for unit pack boxes. The unit pack box shall additionally be labeled in red: “UNIT PACK BOX. DO NOT OPEN EXCEPT FOR USE OR OTHERWISE DIRECTED.”

3.2.6.2.6.5 Overpacking and marking unit packs. When directed by the GPA, the unit pack boxes going to the same destination shall be overpacked into an outer container meeting the requirements of 3.1.1.1, 3.1.1.2 or 3.1.1.3 and marked in accordance with 3.3.3. The number of unit pack boxes packaged into each overpack container shall be in accordance with the GPA provided shipping paperwork (DD Form 250 or DD Form 1149). Loose fill material is not allowed.

3.2.7 Special tools and spare components. Special tools, which are not required to be installed in the equipment, shall be separately packaged and packed as spare components. Spare components including On-Board Repair Parts (OBRP), Shipyard Load Lists (SLL) Items, Initial Navy Supply System Repair Parts (INSSRP), items bought to Individual Repair Part Ordering Data (IRPOD) and Field Change (F/C) kit material shall be packaged, packed and marked as follows: ESD control requirements are defined in 3.1.1.7, ESD handling requirements are defined in 3.2.7.1.4.4, and moisture protection requirements are defined in 3.2.7.1.6.

3.2.7.1 Unit packaging quantity. Unless otherwise specified by the GPA, all items shall be individually unit packaged. Each unit package shall conform and be labeled to requirements noted elsewhere in this specification. A unit is defined to be a single item, set, box, kit, assembly or as specified by the GPA. For surface mount components, each reel, tray, or tube shall be considered a single unit (see 3.2.6.2). All items included in a set/kit of OBRP, INS, INSSRP or SLL shall be individually unit packaged.

3.2.7.1.1 Bags. Bag size should be kept to a minimum. Prior to sealing the bag, excess atmosphere shall be expelled by reducing the bag to its smallest volume practical without damage to the component. Bag materials shall meet the requirements of 3.1.1.7, 3.1.1.8 and/or 3.1.1.9 as applicable. Bag size is based on the inside dimensions.

3.2.7.1.2 Parts/assemblies under six inches in length, width or depth. This bag should be a standard bag three inches by three inches (or smaller) whenever possible. The size of the

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bag should not exceed two inches in height or width greater than the part/assembly after the part/assembly has been placed inside the bag. However, to allow space for label information for bags containing very small parts, the Packager/Seller is not required to utilize a bag smaller than three inches in height or three inches in width. The Seller may increase the bag size due to GPA/Government supplied labels as specified in 3.2.9.3 below.

3.2.7.1.2.1 Resealing requirements. For sealed unit pack bags containing parts or assemblies less than six inches in length, width or depth, the bag shall permit opening and resealing once. All other bags shall permit opening and resealing two times. The intent is to provide sufficient material so that the bag may be opened by cutting off the portion that is sealed and then resealed using a heat sealer.

3.2.7.1.3 Parts/assemblies greater or equal to six inches in length, width or depth. The size of the bag should not exceed four inches in height or width greater than the part/assembly size after the part/assembly has been placed inside the bag. The bag shall permit opening and resealing two times. The intent is to provide sufficient material so that the bag may be opened by cutting off the portion that is sealed and then resealed using a heat sealer.

3.2.7.1.4 Cushioning and protection.

3.2.7.1.4.1 Unit-pack containers (boxes, fastpacks, totes, trays, tubes, vials, etc.). The use of unit pack containers for unit packaging is permitted, provided it provides packaging that is simpler, less costly, and complies with the requirements stated herein.

3.2.7.1.4.2 Cushioning material thickness. The cushioning for the protection of the leads and terminals shall be sufficient to prevent physical damage and maintain leads and terminals in the manufactured condition under handling and transportation environments.

3.2.7.1.4.3 ESD cushioning. For ESD sensitive multi-pin electronic devices, the cushioning material shall be high-density conductive foam.

3.2.7.1.4.4 ESD handling. All ESD sensitive components shall be handled and packaged only at a field force protected workstation. For packaging purposes, all microcircuits, and any part marked by the part manufacturer with one or more triangles to identify the part is static sensitive, are considered to be ESD sensitive electronic devices.

3.2.7.1.5 Cushioning and protection (other than as discussed in 3.2.7.1.4). Deleted.

3.2.7.1.6 Protection from moisture. Barrier material per 3.1.1.7 shall be used. The required desiccant and humidity indicator shall be attached to the outside of an inner bag or wrap. The desiccant shall not come in contact with the item being packaged. The reseal requirements identified in paragraph 3.2.7.1.2.1 of this specification apply to the unit package bag. Unit package marking is only required on the outer bag. In addition to marking requirements specified elsewhere in this specification, the unit packaging shall be marked “DESICCATED PACKAGE - DO NOT OPEN UNTIL READY FOR USE”.

Barrier bag material for all plastic encapsulated microcircuits (PEMs) shall be in accordance with 3.1.1.9.

3.2.7.2 Unit pack. Spare components shall be individually packed unless the provisions of 3.2.10 apply.

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3.2.7.3 Marking. See 3.2.9, 3.2.10 and 3.3.1.

3.2.8 Circuit breaker packaging, packing and marking. Circuit breakers designed and manufactured in accordance with the Circuit Breaker Military Specification (MIL-DTL-17361 (SH) for AQB type and MIL-DTL-17587(SH) for ACB type circuit breakers) shall be preserved, packaged, packed, and marked per the requirements of MIL-STD-2073-1 Method 50 (waterproof and vaporproof protection with desiccant and humidity indicator). Materials shall be in accordance with 3.1. Marking shall be in accordance to MIL-STD-129. Each circuit breaker shall be blocked and braced to prevent movement within the shipping container, and each shipping container shall be secured to the shipping pallet using metal bands.

3.2.9 Unit pack labels.

3.2.9.1 Warranty markings. Warranty markings are not required on sealed unit pack bags.

3.2.9.2 Marking size and font. The marking size and font, including bar code density, shall be determined such that the label/marking requirements will not increase the package size. Except as noted in 3.2.9.3, the package size shall be determined per 3.2.7.1.2 and not by the label/marking size. All markings must meet the requirements of MIL-STD-129 except as modified herein. If a marking requirement exists in MIL-STD-129, but is not specifically included herein, it shall not apply unless specified by the GPA.

3.2.9.3 GPA/Government supplied labels. Labels fabricated by the Packager/Seller shall conform to 3.2.9.2 above. Any label, supplied by the GPA or a Government Agency, which exceeds the size of the bag (as specified in 3.2.7.1.2) may be trimmed and/or folded over both sides of the bag provided the label remains legible and bar codes are not folded. If the Packager/Seller is unable to fold and trim the label to meet the requirements of 3.2.7.1.2, then a larger bag may be used. In this case, the bag size shall be determined based on the size of the folded and trimmed label and the part size.

3.2.10 Field change kits.

3.2.10.1 Packaging. Items being furnished as part of a field change kit shall be packaged in accordance with the applicable requirements specified herein, for the type of item being shipped (e.g., 3.2.2 for drawers and chassis; 3.2.6 for miscellaneous items; and 3.2.7 for spare components, etc.). Packaging instructions provided by the GPA take precedence over the requirements stated herein.

Similar items shall be packaged as a lot provided the lot size does not exceed the number of parts required to modify one assembly. For example, if four identical screws are required to modify a given drawer and there are two drawers per ship, then the field change kit should allow four screws per bag. Spare items shall be packaged to allow for shipment separate from the installed items.

3.2.10.2 Packing. Items being furnished as part of a field change kit shall be packed in accordance with the applicable requirements specified herein, for the type of item being shipped (e.g., 3.2.2 for drawers and chassis, 3.2.6 for miscellaneous items, and 3.2.7 for spare components, etc.).

If the GPA has contracted for the Seller to install the field change kit and long-term storage is

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not involved, all items (except for spare parts) may be over-packed in one or more wooden or fiberboard boxes per the applicable requirements herein. Alternate methods of over-packing will be considered and authorized by the GPA on a case-by-case basis. (For example, over packing PWAs in plastic cases or combining the entire unit packaged material in aluminum or wood shipping crates.) Spare parts shall be packed in a separate container in accordance with 3.2.7.

3.2.10.3 Marking. Marking shall be in accordance with 3.3.

3.2.11 Heat Exchangers, cold plates and other similar items. Heat exchangers, cold plates and other similar designed items, that are being shipped as spare components, shall be PP&M the same as for drawers or chassis (3.2.2). Each item shall be packed in a separate wooden box. A wooden box is not required for modular electronic equipment with integral heat sinks (e.g., the Generic I&C power supply assemblies); these items are covered by 3.2.6.1.

3.2.11.1 Prior to packaging. Prior to packaging, each item shall be purged free of water using clean, dry shop air. Subsequently, the inlet and outlet ports shall be capped with a suitable cap (of a commercial material that is not detrimental to the part being capped and is not polyvinyl chloride) to preclude the entrance of moisture or foreign material. The caps/plugs shall be designed such that they cannot be inadvertently pushed into the piping ports. This requirement also applies to items that are being shipped installed in a cabinet, enclosure or panel.

3.2.11.2 Appropriate cushioning material and support. Appropriate cushioning material/support shall be provided to preclude damage, bending, deforming, or otherwise causing a dimensional change to the orientation of the inlet port, outlet port, piping and heat exchanger fins. This requirement also applies to items that are being shipped installed in a cabinet, enclosure or panel.

3.2.11.3 Desiccants and humidity cards. Desiccants and humidity cards are not required.

3.2.11.4 Unit package label. A label shall be provided with each unit package that contains the following marking, or equivalent:

CAUTION:

DO NOT REMOVE CAPS FROM THE PIPING INLET AND OUTLET PORTS EXCEPT FOR INSPECTION OR INSTALLATION. CAPS MUST BE REMOVED PRIOR TO INSTALLATION.

This label shall also be provided on the final pack for items installed in a cabinet, enclosure or panel.

3.2.12 Dense, heavy, irregular loads. When the load is characterized by irregular shape not lending support to the container and/or by great density (e.g., Converter bus planes and bus bar assemblies, large heavy transformers, centrifugal air blowers, and gamma ray shields). Dense, heavy, irregular loads must be wrapped in A-A-3174 plastic sheet, then blocked, braced and anchored in MIL-PRF-83671, Class 1, Rigid Foam to prevent movement of the load inside the box. Dense, heavy, irregular loads must be packed in wooden boxes (3.1.1.3) if the load is less than 800 pounds or wooden crates (3.1.1.4) if the load is greater than 800 pounds. The wooden box or crate shall be designed to be shipped and handled under cover and stored in warehouses or other structures affording equivalent protection from weather. The packing shall

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afford adequate protection during multiple reshipments.

3.3 Marking requirements. All markings must meet the applicable requirements of MIL-STD-129 except as modified herein. If a marking requirement exists in MIL-STD-129, but is not specifically included herein, it shall not apply unless specified by the GPA.

3.3.1 Unit pack marking.

3.3.1.1 Minimum label information. The following minimum information to be printed on the label shall include:

(a) Bar Code Marking
(b) National Stock Number (See note below)
(c) Description
(d) Drawing/Piece Number, MIL Designation or Catalog Number
(e) Quantity and Unit of Measure
(f) Purchase Order Number
(g) Supplier Name or CAGE Code
(h) Equipment Line Item Number (ELIN) when supplied by the GPA
(i) Serial Nmber

Note: National Stock Number (NSN) shall deviate from MIL-STD-129 marking requirements. MIL-STD-129 requires a 13 digit NSN, whereas the GPA requires a 17 character NSN. The 17 character NSN includes: (First two digits is the Cognizant Code, next four digits is the Federal Supply Class, next nine digits is the National Item Identification Number, and the last two digits is the Special Material Identification Code.)

3.3.1.1.1 Labels for ESD sensitive components. In addition to the above markings, all labels for ESD sensitive components shall be marked with the standard ESD symbol per MIL-STD-129 and the statement “DO NOT OPEN EXCEPT AT APPROVED FIELD FORCE PROTECTIVE WORK STATION”. The minimum size of the symbol shall be 0.33 inches measured vertically. The historical ESD symbol (three arrows in a circle) should no longer be used.

3.3.1.2 Shelf life marking. When the item being shipped has a finite shelf life (e.g., 60 months or less), insert the following additional information (this requirement applies to the end item and not to elastomers installed in components):

(a) Type I Shelf Life Item
(b) Cure Date (Seller to insert cure date if an elastomer)
(c) Expires (Seller to insert shelf life expiration date)
(d) To be disposed of upon expiration

3.3.1.2.1 Elastomer shelf life note. Elastomers with a shelf life between five and 19 years must be marked with a shelf life of five years. Elastomers with a shelf life of 20 years or greater must be marked “Cure Date____. Do not use 20 years beyond cure date.”

3.3.2 Intermediate packing and marking. Intermediate packing for field change kit material shall be in accordance with 3.2.10 and as specified in the field change ordering data. Individually packaged components may be over packed in a temporary shipping container for transportation to the destination.

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3.3.3 Outer container marking. Exterior markings shall contain only that information specified in 3.3.3.1-4, 6-10, 15 and 16 (mandatory) and 3.3.3.5, 11-14 and 17 (as appropriate) below. Markings on exterior containers shall not reference any: (1) Project, (2) Equipment Specification, or (3) Number of Installed Components.

3.3.3.1 Name and address of Seller. Include the name and address of the Seller.

3.3.3.2 Destination address. Include the complete destination address as provided by GPA and note “For NMS Stock”, ”On-Board-Spares”, ”Shipyard Installation Tools”, ”Tender Load List”, or Tender Allowance List” if applicable. Include hull number. The specific destination will be furnished by the GPA prior to shipment.

3.3.3.3 Item description(s) including National Stock Numbers (NSN) with bar code marking. When shipments include components intended for both installation and on-board spares, the item descriptions should delineate the quantities intended for each use. The item description(s) are not to reference any project or core designation or applicable equipment specifications. When shipments include components intended for both installation and on-board spares, the item description shall delineate the quantities intended for each use. However, no mention shall be made of the quantity of installed components which constitutes a shipset.

3.3.3.4 Standard identifier and serial number. Standard identifier and serial number as provided by the GPA. The serial number must match exactly the serial number marked on the equipment nameplate and shipping document.

3.3.3.5 Equipment line item number. Equipment line item number (ELIN), when supplied by the GPA.

3.3.3.6 Purchase order number. Purchase order number, when supplied by the GPA.

3.3.3.7 Bill of lading number. Government or commercial bill of lading number/TCN number/registered number/way bill number.

3.3.3.8 Gross weight and cube of parcel. Gross weight and cube of the parcel as shipped.

3.3.3.9 Boxes containing field change kits. Boxes containing field change kits shall be identified with the following external markings:

ShipAlt Number_____	Note a.
Field Change Number_____	Note b.
Box ___ of ___	Note c.
TCN/Reg. Number/GBL Way Bill Number_____	Note d.
Ship Hull Number_____	Note e.
Job Material List (JML) Number	Note f.

Notes:

(a) To be supplied from the GPA if available. If not available, mark “Not Available” or “Later” as directed by GPA.

(b) To be supplied by GPA. Do not include project identifier (i.e. S6W).

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(c) Mark each box “1 of 3", “2 of 3", “3 of 3" etc. If shipment is made from a single box, mark it 1 of 1". Only complete kits shall be shipped unless GPA specifically authorizes shipments of partial kits; in such cases, GPA may provide additional marking instructions to suitably indicate the shortage.

(d) Enter data as applicable to the shipment.

(e) To be supplied from the GPA if available. If not available, then mark “Not Available” or “Later” as directed by GPA.

(f) JML numbers shall be affixed to each package when the field change kits are shipping directly to a naval shipyard. This information will be supplied by GPA, when applicable.

3.3.3.10 Precautionary note or label for Government-Furnished Equipment. A precautionary note or label (must be in red): “Government-Furnished Equipment - Open per Requirements of Cognizant Government Inspector”.

3.3.3.11 Precautionary note or label for warehouse storage only. A precautionary note or label (must be in red): “Warehouse Storage Only”.

3.3.3.12 Primary plant detectors. For primary plant detectors, the container or unpacked item shall bear the precaution note (must be in red): “Do Not Open Seal Closures Except in Clean Area”.

3.3.3.13 Container “UP” arrow markings. If arrows are required to indicate the desired position of the container during shipment and storage, place an arrow and the word “UP” on all four sides of the container.

3.3.3.14 Forklift markings. If applicable, mark “FORKLIFT HERE”.

3.3.3.15 Packing list. Two copies of the DD Form 250/DD Form 1149 shall be placed in a waterproof envelope labeled “Packing List” attached to the outside of each container. In addition, one copy of the DD Form 250/DD Form 1149 shall be placed in a waterproof envelope attached to the inside of the container.

3.3.3.15.1 Government inspection and handling shipping forms. When Government inspection is required, the Government inspector may elect to sign only the shipping forms that are attached to the outside of each container. If this option is chosen, the Seller shall add a note to the form provided inside the container that indicates that forms signed by the Government Inspector are included in the envelope attached to the outside of the container.

3.3.3.15.2 Security requirements. When multiple containers are used for a sensitive unclassified or classified shipment of equipment, the security requirements on the purchase order shall apply.

3.3.3.16 Center of balance marking. Mark the center of balance as required by MIL-STD-129 and 3.2.1.3.2 herein.

3.3.3.17 Single or multiple box marking. If the shipment consists of one box, mark it “1” of “1”. If multiple boxes are involved, each box shall be marked “1 of 5”, “2 of 5”, etc.

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3.3.3.18 Marking of over-pack containers containing multiple unit pack items. If the shipment consists of two or more unit packed items that are over-packed in an outer over-pack container, then the over-pack container shall be marked per 3.3.1.1 (for all items in the over-pack).

3.4 Additional requirements applicable to items that are assigned 2S/X1 National Stock Numbers.

3.4.1 Packaging for all 2S/X1 components. Note: 2S/X1 NSNs are generally assigned to major components for which the GPA requires nameplates/serialization. For all 2S/X1 components, the unit pack and exterior container must be one and the same (i.e., each 2S/X1 component must be individually packaged for shipment). Individually packaged components may be overpacked in a temporary shipping container for transportation to the destination. The overpack container shall not be marked with the GPA provided bar coded labels. Overpack markings shall be in accordance with applicable sections of MIL-STD-129, and any exceptions noted below.

3.4.2 Bar coded labels for 2S/X1 items. In conjunction with providing shipping destinations and applicable shipping documents, upon Seller request, the GPA will supply for Seller use two (2) bar coded labels for each component that is assigned a 2S/X1 NSN. The Seller shall affix one of these labels immediately above the exterior markings required above. The second GPA furnished label shall be affixed completely opposite of the first (i.e., if the first label is on the front upper left, the second label is on back panel lower right).

3.4.3 Tags used for exterior marking requirements. When tags are used to meet the requirements for exterior markings the tags shall be made of plastic in accordance with applicable MIL-STD-129 requirements, and the bar coded labels shall be applied to the front (preferred) or reverse side of such tags.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the Seller is responsible for the performance of inspection requirements as follows. If PP&M is performed at a Packaging Supplier and the equipment is to be drop shipped from the Packaging Supplier, the Seller shall perform periodic audits and over checks to determine compliance with this specification and the applicable PP&M procedure.

(a) PP&M shall be visually inspected as necessary to determine compliance with this specification. Sample inspection per the Seller's sample inspection procedure is acceptable.

(b) The shipping vehicle shall be visually inspected when the shipment exceeds 1,000 pounds. Shipping vehicles with unsatisfactory conditions shall not be used. All truck shipments shall have a driver vehicle inspection report in accordance with DOT CFR Title 49, Chapter III, Part 396.11 "Inspection, Repair, and Maintenance". The items to be checked on the loaded vehicle prior to release shall be identified in the supplier's loading and blocking procedure (see 1.3.1) or the packaging, packing, and shipping procedure when the shipment exceeds 1,000 pounds. As a minimum, the loaded truck shall be inspected for (1) load shored and tied down as needed, (2) weight properly distributed and not overloaded, and (3) tailgates and doors closed, and all equipment secured.

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(c) For surface mount devices (SMD) packaged on reels (tape and reel), see paragraph 3.2.6.2.2.1 sample inspection requirements.

5. PREPARATION FOR DELIVERY

5.1 Requirements for classified shipments. Classified shipments shall be in accordance with the National Industrial Security Program Operating Manual (NISPOM). Seller shall verify that truck driver has United States citizenship per NISPOM verification requirements. Additional classified hardware markings shall be as specified in the latest revision of the Contract Security Classification Specification and its attachments.

5.2 Exclusive use of truck. When shipment by "exclusive use of truck" is specified by the GPA, the expectation is that the trailer and the tractor (including the dromedary, if applicable) shall contain only those items that have been authorized for shipment by the GPA. Tandem trailers are also not allowed, except for shipment of items authorized by the GPA.

5.3 Explosives and other hazardous materials. No equipment shall be shipped in any transportation vehicle that contains explosive or hazardous materials (except for those materials specifically authorized by the GPA).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions and acronyms.

CCA	Circuit Card Assembly
DOT	Department of Transportation
ESD	Electrostatic Discharge
F/C	Field Change
GBL	Government Bill of Lading
GPA	Government Procurement Agency
INSSRP	Initial Navy Supply System Repair Parts
INS	Initial Navy Stock
JML	Job Material List
Marking	Marking, as used in this specification refers to marking and labeling of packaging and packing.
MBB	Moisture Barrier Bags
NMS	NAVSEA Managed Spares (Spare Equipment)
NSN	National Stock Number
OBRP	On-Board Repair Parts
OBS	On-Board Spares
PEM	Plastic Encapsulated Microcircuit
PP&M	Packaging, Packing and Marking
PWB	Printed Wiring Board (Printed Board)
PWA	Printed Wiring Assembly
SMD	Surface Mount Device
SLL	Shipyard Load List
TCN	Transportation Control Number
WVTR	Water Vapor Transmission Rate

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6.2 Changes from previous issue: marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing Activity:
Navy – SH
Project PACK-2023-005

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